

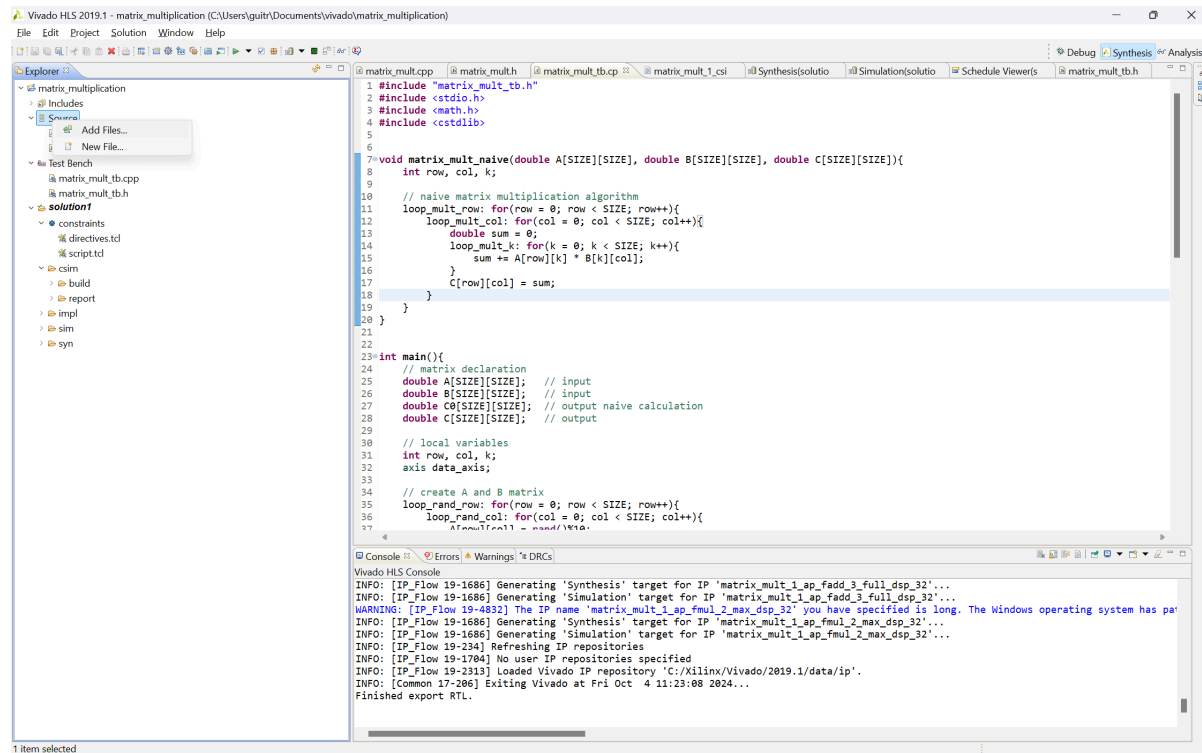
Vivado HLS 2019.1 - Create Project

In Vivado HLS 2019.1 run the following:

Add/Create Files

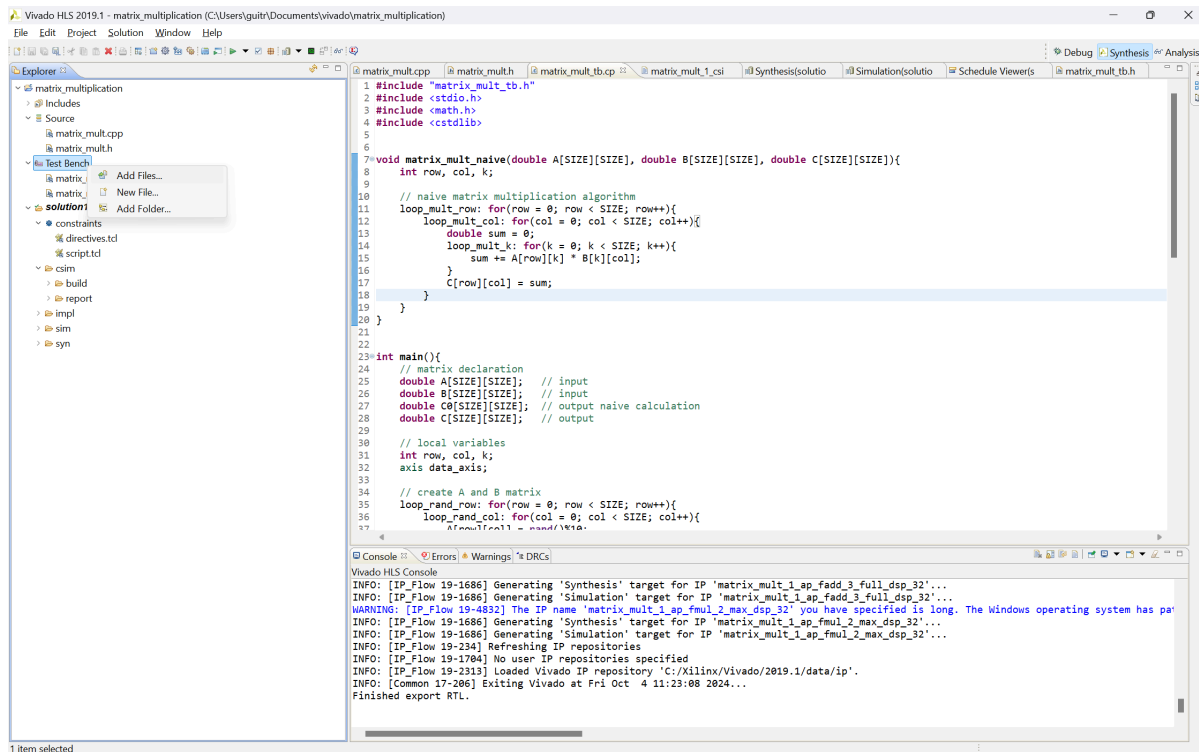
Source

Both existing and new files may be created by right mouse click and **Add Files...** or **New Files...**:



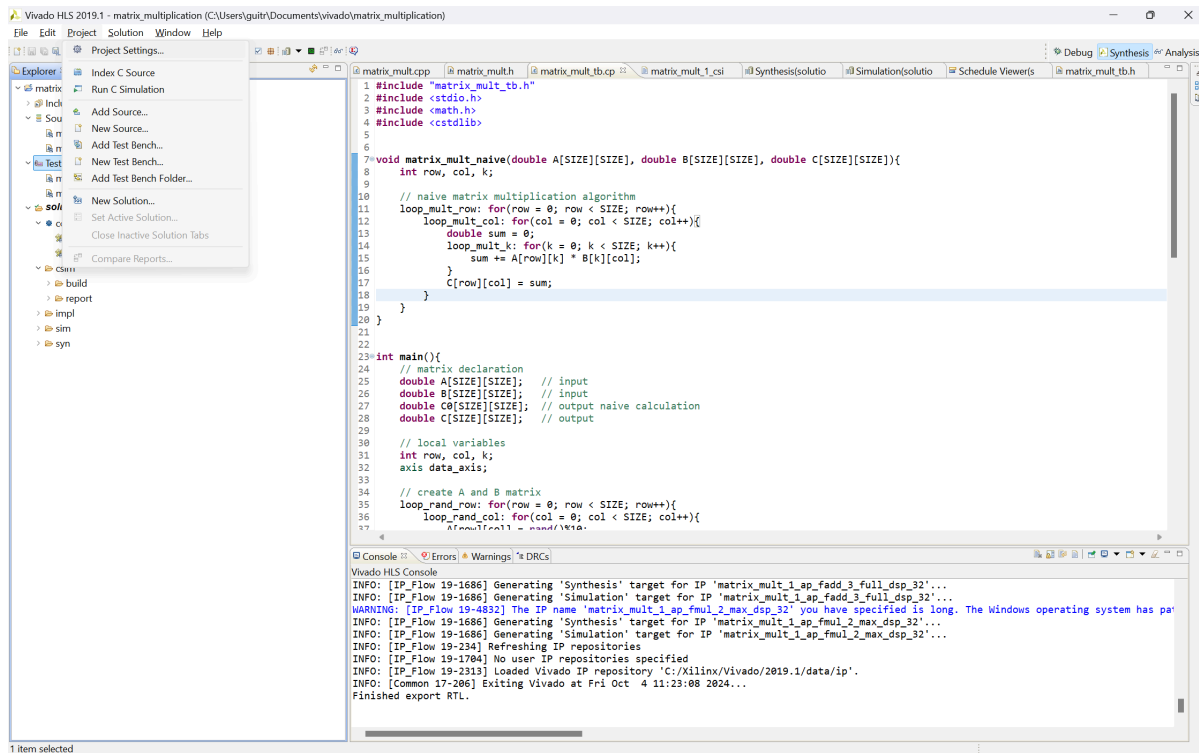
Test Bench

Both existing and files may be created by right mouse click and Add Files... or New Files...:

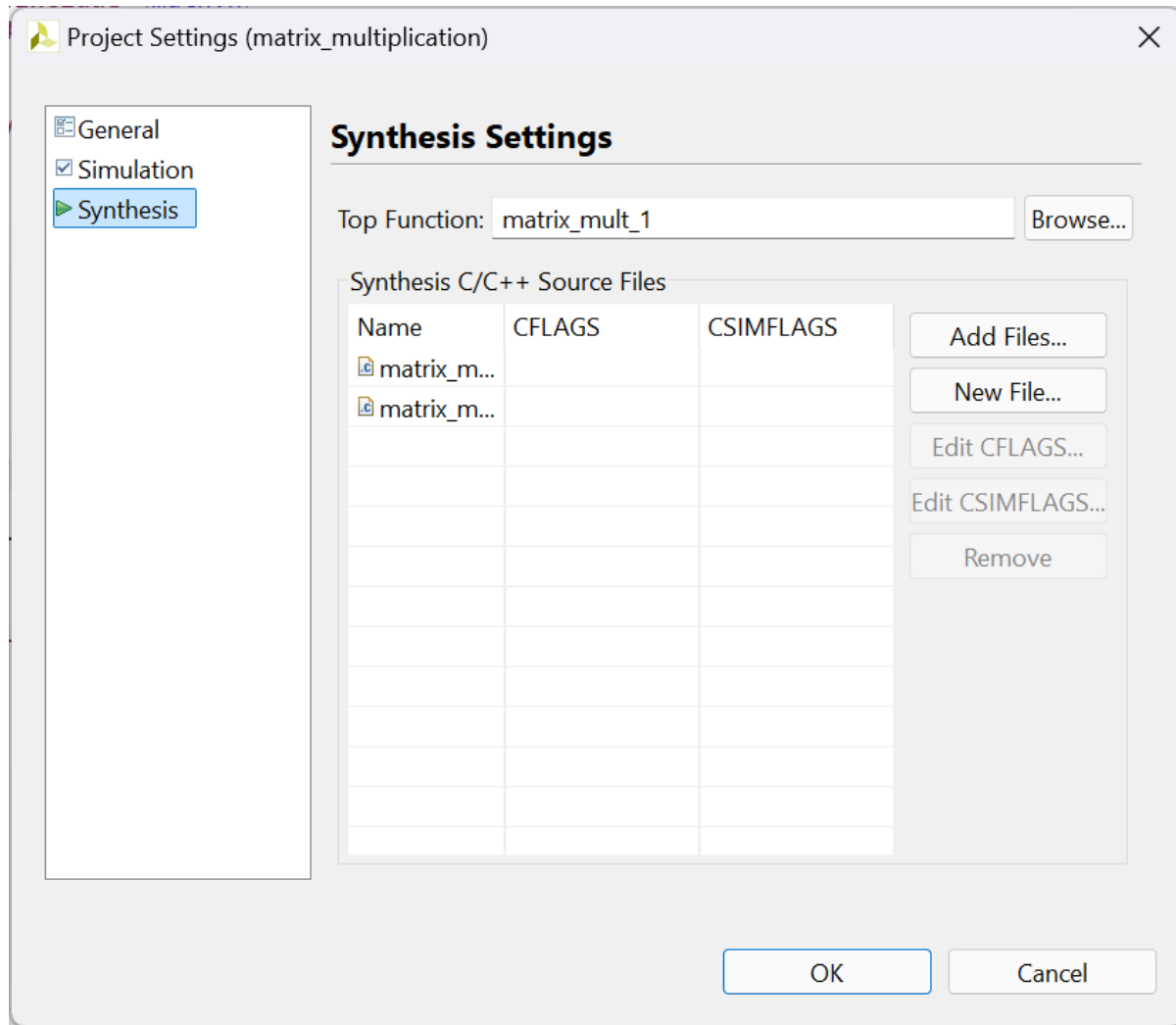


Project Settings

After adding the necessary files and functions it is necessary to modify the Project Settings to consider the functions desired to Synthesize.



Under Top Function include the main function to be Synthesize:



Create Solution

Vivado HLS 2019.1 - matrix_multiplication (C:/Users/guith/Documents/vivado/matrix_multiplication)

File Edit Project Solution Window Help

Explorer

matrix_multiplication

Includes

Source

matrix_mult.cpp

matrix_mult.h

Test Bench

matrix_mult_tb.c

matrix_mult_tb.h

solution0

solution1

matrix_mult.cpp

1#include <vector>h

2#include <hls_stream.h>

3#include <ap_int.h>

4

5#define SIZE 8 // matrix size

6#define solution0 // active solution

7

8// define matrix data type

9#ifdef solution0

10typedef int matrix;

11#endif

12

13#ifdef solution1

14typedef float matrix;

15#endif

16

17// define axis data structure

18struct axis{

19matrix data;

20ap_uint<1> last;

21};

22

23#ifdef solution0

24void matrix_mult_0(hls::stream<axis> &input_stream, hls::stream<axis> &output_stream);

25#endif

26

27#ifdef solution1

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Outline

Directives

1std.h

2hls_stream.h

3ap_int.h

4SIZE

5solution0

6matrix: int

7matrix: float

8axis

9matrix_mult_0(hls::stream<axis> &hls::stream<axis> &): void

10matrix_mult_1(hls::stream<axis> &hls::stream<axis> &): void

Console

Errors

Warnings

W DRCs

Vivado HLS Console

source run_ippack.tcl -notrace

create_project: Time (s): cpu = 00:00:01 ; elapsed = 00:00:06 . Memory (MB): peak = 357.398 ; gain = 59.184

INFO: [IP_Flow 19-234] Refreshing IP repositories

INFO: [IP_Flow 19-1704] No user IP repositories specified

INFO: [IP_Flow 19-2313] Loaded Vivado IP repository 'C:/Xilinx/Vivado/2019.1/data/ip'.

WARNING: [IP_Flow 19-4832] The IP name 'matrix_mult_1_ap_fadd_3_full_dsp_32' you have specified is long. The Windows operating system has path length limitations. It is recommended to use a shorter name.

create_ip: Time (s): cpu = 00:00:03 ; elapsed = 00:00:07 . Memory (MB): peak = 358.001 ; gain = 41.402

INFO: [IP_Flow 19-1686] Generating 'Synthesis' target for IP 'matrix_mult_1_ap_fadd_3_full_dsp_32'...

INFO: [IP_Flow 19-1686] Generating 'Simulation' target for IP 'matrix_mult_1_ap_fadd_3_full_dsp_32'...

WARNING: [IP_Flow 19-4832] The IP name 'matrix_mult_1_ap_fmuls_2_max_dsp_32' you have specified is long. The Windows operating system has path length limitations. It is recommended to use a shorter name.

INFO: [IP_Flow 19-1686] Generating 'Synthesis' target for IP 'matrix_mult_1_ap_fmuls_2_max_dsp_32'...

INFO: [IP_Flow 19-1686] Generating 'Simulation' target for IP 'matrix_mult_1_ap_fmuls_2_max_dsp_32'...

INFO: [IP_Flow 19-234] Refreshing IP repositories

INFO: [IP_Flow 19-1704] No user IP repositories specified

INFO: [IP_Flow 19-2313] Loaded Vivado IP repository 'C:/Xilinx/Vivado/2019.1/data/ip'.

INFO: [Common 17-206] Exiting Vivado at Tue Oct 8 10:57:03 2024...

Finished export RTL.

Starting C simulation ...

C:/Xilinx/Vivado/2019.1/bin/vivado_hls.bat C:/Users/guith/Documents/vivado/matrix_multiplication/solution1/csim.tcl

INFO: [HLS 200-10] Running 'C:/Xilinx/Vivado/2019.1/bin/unwrapped/win64.o/vivado_hls.exe'

Writable

Smart Insert

14:22

Solution Configuration



Create Vivado HLS solution for selected technology

Solution Name:

Clock

Period: Uncertainty:

Part Selection

Part: **xc7z020clg484-1**



Options

☒ Copy directives and constraints from solution:

☐ SDAccel Bottom Up Flow

Finish

Cancel

Solution Configuration



Create Vivado HLS solution for selected technology

Solution Name:

Clock

Period: Uncertainty:

Part Selection

Part: **ZedBoard Zynq Evaluation and Development Kit (p**

Options

☐ Copy directives and constraints from solution: ▾

☐ SDAccel Bottom Up Flow

Select this to generate a SDAccel ke

Finish

Cancel

Vivado HLS 2019.1 - matrix_multiplication (C:/Users/guith/Documents/vivado/matrix_multiplication)

File Edit Project Solution Window Help

Explorer

matrix_multiplication
└─ Includes
└─ Source
└─ matrix_mult.cpp
└─ matrix_mult.h
└─ Test Bench
└─ matrix_mult_tb.c
└─ matrix_mult_tb.h
└─ solution0
└─ solution1

matrix_mult Solution Settings... .cpp

matrix_mult_tb.h

matrix_mult_tb.cpp

matrix_mult.h

```
1 #include <stdio.h>
2 #include <hls_stream.h>
3 #include <ap_int.h>
4
5 #define SIZE 8 // matrix size
6 #define solution0 // active solution
7
8 // define matrix data type
9 #ifdef solution0
10 typedef int matrix;
11 #endif
12
13 #ifdef solution1
14 typedef float matrix;
15 #endif
16
17 // define axis data structure
18 struct axis{
19     matrix data;
20     ap_uint<1> last;
21 };
22
23 #ifdef solution0
24 void matrix_mult_0(hls::stream<axis> &input_stream, hls::stream<axis> &output_stream);
25 #endif
26
27 #ifdef solution1
```

Outline

Directive

1 stdio.h

2 hls_stream.h

3 ap_int.h

4 SIZE

5 solution0

6 matrix: int

7 matrix: float

8 axis

9 matrix_mult_0(hls::stream<axis>&8, hls::stream<axis>&8) : void

10 matrix_mult_1(hls::stream<axis>&8, hls::stream<axis>&8) : void

Console

Errors

Warnings: 1x DRCs

Vivado HLS Console

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C:/Xilinx/Vivado/2019.1/bin/vivado_hls.bat C:/Users/guith/Documents/vivado/matrix_multiplication/solution1/csim.tcl
INFO: [HLS 200-10] Running 'C:/Xilinx/Vivado/2019.1/bin/unwrapped/win64.o/vivado_hls.exe'
```

Writable Smart Insert 14:22